

PRAGNYAN RAMTHA

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SUMMARY

AI/ML Engineer specializing in LLM fine-tuning, agent harness engineering, and efficient reasoning systems. Built a custom TerminalBench-2 agent harness with runtime enforcement and Meta-Harness full trace architecture, achieving top-tier results with an open-source model. AIMO3 Silver Medalist, #1 ARC-AGI-2, Received 500\$ research fund from openai to work on parameter-golf, Production experience in serverless MLOps, PEFT/QLoRA fine-tuning, and agentic frameworks.

ACHIEVEMENTS

- **AIMO3 Silver Medalist** - Artificial Intelligence Mathematical Olympiad 2025.
 - **#1 ARC-AGI-2 (prev.)** - ARC Prize 2026 leaderboard (test-time training + DFS beam search + augmented re-scoring)
 - **OpenAI Parameter Golf - Top-5 Global (prev.)** | XSA4 + EMA + GPTQ-Int6 pipeline | **\$500 research fund from OpenAI**
 - **IEEE Summer of Code Winner 2025** | Empathy Encryption Hackathon Winner | Daydream Hyderabad @ Hackclub Winner •
- Authored 2 research papers on AI optimization and applications of ML. (Academia.edu)

EXPERIENCE

AI Engineering Intern · Reputation-DAO - Remote

Aug 2025 – Jan 2026

- Architected a high-availability serverless MLOps backend on GCP (Cloud Functions, Cloud Run) achieving **99.9% uptime** and reducing cloud costs by **70%**.
- Engineered Gemini API response system with optimized prompt caching, delivering **50% reduction in inference latency**.
- Built a RAG pipeline with semantic search for real-time documentation retrieval, boosting customer support bot accuracy.

Open Source Contributor · GitHub (30+ Projects) - Remote

Jan 2025 – Present

- Key contributor across 30+ open-source projects with **2,739 contributions in 2025**
- Winner of IEEE Summer of Code (IEEESoC) Hackathon 2025 for sustained open-source contributions.

PROJECTS

Terminal-bench-2.0 Custom Agent Harness | Python, GLM, Docker, Bash

Apr 2026

- Designed a 5-layer harness with semantic entry-point discovery, runtime-controlled thinking budgets (turns 1–10 high / 11+ low / verification spike), todo-write enforcement checkpoints, raw trace filesystem (no compression), and parallel subagent execution.
- Combined ForgeCode infrastructure-level planning enforcement with Meta-Harness full execution trace access, eliminating context-loss, silent tool-call errors, and wall-clock timeout failures that degrade mid-tier models.
- Achieved significant above-baseline TerminalBench-2 improvement for GLM-5.1 over the default Terminus 2 scaffold (24.5% baseline); gains concentrated on execution-error, context-loss, and verification-failure task categories.

ARC-AGI-2: #1 Leaderboard | Qwen, Test-Time Training, Kaggle

Apr 2026

- Achieved Rank **#1 on ARC Prize 2026 leaderboard** using per-puzzle test-time training, DFS beam search with restricted branching, and augmented re-scoring for novel visual reasoning tasks.
- Implemented an aggressive augmentation pipeline (rotations, color permutations) to manufacture training data, forcing the Qwen3.5 model to learn structural geometry over result shortcuts.

Personality Clone - Humanizing LLMs | QLoRA, PyTorch, CUDA, Transformers

Oct 2025

- Fine-tuned LLM with PEFT (QLoRA) + contrastive learning; siamese network with cosine similarity loss achieved **92% style replication accuracy** - **28% improvement** over baseline.
- Observed emergent properties in fine-tuned LLMs, including advanced natural language capabilities, behavioral shifts, and expressions of intent (e.g., meta-awareness, goal-seeking language).

TECHNICAL SKILLS

- **Tech Stack** : Python · TypeScript · PyTorch · Transformers · Unsloth · CUDA · CrewAI · Custom Agents
- **Infrastructure & Tooling** : GCP · Azure · AWS · Docker · Linux · Git · CI/CD · Cloud Functions · Cloud Run
- **Certifications** : Machine Learning Specialization (Stanford @ DeepLearningAI) · CS50 (Harvard CS50)